SDG 7 — Affordable and Clean Energy

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Why we chose SDG 7

Of the 17 Sustainable Development Goals we selected **SDG 7: "Ensure access to affordable, reliable, sustainable and modern energy for all"**. Energy sits underneath almost every other goal (health, education, industry, climate action), and it is also the goal where embedded and connected systems can make the most direct, measurable difference: every kilowatt-hour that a sensor-driven controller avoids wasting is a kilowatt-hour that does not have to be generated. That makes SDG 7 a natural fit for a lab project in Computational Models for Embedded Systems.

The targets

- **7.1 — Universal access** to electricity (7.1.1) and to clean cooking fuels and technologies (7.1.2).
- **7.2 — Renewables:** substantially increase the share of renewable energy in the global energy mix.
- **7.3 — Efficiency:** double the global rate of improvement in energy intensity by 2030, compared with the 1990–2010 average.
- **7.a / 7.b — Means of implementation:** international cooperation, investment and infrastructure for clean energy, especially in developing countries.

Where the world stands (Tracking SDG 7 — Energy Progress Report 2025)

- **Access:** about 92% of the world's population had electricity in 2023 (87% in 2010), but **666 million people** still have none, most of them in rural sub-Saharan Africa [1, 2].
- **Clean cooking:** about **2.1 billion people** still cook with polluting fuels; progress is outpaced by population growth, and on current trends only 78% will have clean cooking by 2030 [1, 2].
- **Renewables:** only **17.9%** of total final energy consumption came from renewables in 2022 [2].
- **Efficiency:** to still meet target 7.3, energy intensity must now improve by about **4% per year**, well above what has been achieved recently [1].

Main challenges

1. **Unequal access and financing.** The remaining gap is concentrated in poor, rural and fragile regions, where grid extension is expensive and investors see high risk.
2. **Efficiency lags behind renewables.** Policy attention and investment go mostly to new generation, while demand-side waste (buildings, appliances, industry) improves slowly. In the EU, buildings alone account for about 40% of energy consumption and 36% of energy-related greenhouse gas emissions [3].
3. **Intermittency and reliability.** Solar and wind vary over the day and the season, so a higher renewable share needs storage, flexible demand and smarter grids.
4. **Lack of measurement.** Many organisations simply do not know where and when they consume energy, so waste (lights and heating in empty rooms, devices left on) goes unnoticed.

Solutions

- **Decentralised renewables** (mini-grids, off-grid solar) to close the access gap quickly and cheaply in remote areas [2].
- **Energy efficiency first:** insulation, LED lighting, efficient appliances, heat pumps — the cheapest energy is the energy that is not used.
- **Digitalisation and smart buildings:** sensors, meters and controllers that adapt lighting, heating and ventilation to real occupancy, and that shift flexible loads to times of high renewable output.
- **Monitoring and transparency:** fine-grained metering and dashboards turn invisible consumption into data that people and automated systems can act on.

Our angle

Our project targets the **efficiency** side of SDG 7 (target 7.3), at the scale we can actually influence: a university campus. Smart, occupancy-aware control of classroom energy use is a low-cost, replicable measure that also supports the renewable side, because a smaller and more predictable demand is easier to cover with on-site solar power.

Sources. [1] UN DESA, *Progress towards Goal 7* (2025), un.org/en/node/208821 [2] IEA, IRENA, UNSD, World Bank, WHO, *Tracking SDG 7: The Energy Progress Report 2025* [3] European Commission, *In focus: Energy efficiency in buildings* (2020) [4] UN, sdgs.un.org/goals/goal7